Grouped recovery: conditional equivalence and projection limits

RM12 BP-R Stage 3 — methodological review

7 October 2026

Status. M1 authorizes grouped-state qualification, not adoption of the rules examined below. These are reviewer deductions from the supplied Laval–Daganzo source and explicitly conditional algebra. They do not establish a coupled ramp trajectory, control benefit, finite-summary impossibility or useful gradient. No traffic simulation is used.

1 State and sampled motion

At a completed source sample, let x be position (m), v realized speed (m/s), δ time step (s), a_0 zero-speed acceleration (m/s^2), $v_{\max}$ the vehicle capability parameter (m/s), g gravity and $G(x)$ road grade. Fixed lane and type are implicit. The source Appendix A advances

$$x^+ = x + \delta v, \qquad w^+ = v + \delta\{a_0(1 - v/v_{\max}) - gG(x)\}, \tag{1}$$

$$v^+ = \min\{w^+, U^+(x^+)\}. \tag{2}$$

Here w^+ is next desired speed. U^+ denotes the consistently estimated next downstream KW speed, from the model's own coupled field. Equation (2) is not a substitute algorithm for determining that field. At an intermediate phase w^+ cannot be discarded. Birth/node/crossing existence and timing remain separate qualification obligations.

A lane-wise descriptor measure is

$$\mu = \sum_{j=1}^{K} \delta_{\sigma_j}, \qquad \sigma_j = (x_j, v_j, \theta_j, c_j), \qquad \mu^+ = \mathcal{T}_{s\,\#}\mu + \beta_s - \eta_s. \tag{3}$$

θ_j stores fixed motion/type and any required sample-phase data; c_j denotes a cumulative-count anchor (vehicle equivalents), if required by the coupled representation, computed from its own traffic field at the declared moving-boundary sample. It is not a personal vehicle identifier. s is the model's own traffic field, not reference truth; β_s records own-transfer births, and η_s only justified removal. The pushforward maps each signature by the same specified own-state update. This is a conditional description, not a proof that the joined solver is complete. Descriptor count is not physical traffic mass or additional TST population.

2 When a group is lossless

Group only identical *full* signatures, retaining multiplicity m_h:

$$\mu = \sum_{h=1}^{H} m_h \delta_{\sigma_h}, \qquad \mathcal{T}_{s\,\#}\mu = \sum_{h=1}^{H} m_h \delta_{\mathcal{T}_s(\sigma_h)}. \tag{4}$$

This identity preserves the complete measure; $H \le K$, with $H = K$ possible. If the coupled solution is well defined and independent of descriptor names, identical signatures see the same local

field and have identical future updates. Repeated identical constraint inequalities are redundant. Where a qualified cumulative-count construction combines contributions by a minimum,

$$\min\{C_0, C, C, \ldots\} = \min\{C_0, C, \ldots\}. \tag{5}$$

Thus multiplicity does not multiply the service penalty. The road field still contains all continuous transferred mass. These arguments do not merge distinct positions, speeds, count associations or types. Equal current speed alone is insufficient. No assumption is made that sampled births share identical placement, nor that a currently inactive constraint can be permanently removed.

3 Joint location–speed association matters

This is a local-input test of (1)–(2), not a common-rate traffic witness. On a level road, assign illustrative algebraic parameters $\delta = 1$ s, $a_0 = 3$ m/s^2, $v_{\max} = 30$ m/s. Then the desired-speed map is $\psi(v) = 3 + 0.9v$. In two separated regions take the same prescribed local-input speed estimates $U_L = 15$ m/s and $U_R = 20$ m/s. Positions 0 m and 100 m and their one-step advances stay in their respective regions. Compare

$$\mu_A = \delta_{(0,10)} + \delta_{(100,14)}, \qquad \mu_B = \delta_{(0,14)} + \delta_{(100,10)}. \tag{6}$$

They share position and speed marginals, count, mean position and mean speed. Both current speeds are below each assigned local cap. But

$$S_A^+ = \min(12, 15) + \min(15.6, 20) = 27.6 \text{ m/s}, \tag{7}$$
$$S_B^+ = \min(15.6, 15) + \min(12, 20) = 27.0 \text{ m/s}. \tag{8}$$

Separate marginals cannot determine this speed-sum update. A coarse spatial bin covering both regions can lose the same association, even with velocity bins. Resolving the joint support may distinguish it. No claim is made that the prescribed field is the eventual joined solution for both configurations, or that this difference changes exit flow, TST or an optimal action.

4 Count and mean position do not close histogram transport

At one position update, use the source's current-speed advance $x^+ = x + h$ with $h = \delta v = 2$ m and common $v = 2$ m/s, $\delta = 1$ s. All positions are initially in the interval $[0, 10)$ m:

$$A = \{5.5, 7.5\}, \qquad B = \{4.5, 8.5\}. \tag{9}$$

Both have count two, mean 6.5 m and the same speed list. After translation, $A^+ = \{7.5, 9.5\}$ stays inside while $B^+ = \{6.5, 10.5\}$ sends one record across the boundary. Hence their projected next bin counts differ, despite equal initial count/mean summaries. This does not defeat a summary retaining the full subcell profile or a justified richer/restricted family.

For bins B_a and projection $P_a\mu = \mu(B_a)$, the exact formula

$$(P\mathcal{T}_{s\,\#}\mu)_a = \int \mathbf{1}_{B_a}(\mathcal{T}_s(z))\,\mathrm{d}\mu(z) \tag{10}$$

requires the within-bin measure. Calling this a formula does not make it an update from $P\mu$ alone. A declared reconstruction R can instead define a closed approximate map $P\mathcal{T}_sR$, with births/coupling added consistently. Its accuracy, activity and service are separate qualifications. A later bin refinement cannot retrieve lost member locations from the reference.

5 Conditional service and objective bounds

For *already qualified* trajectory contributions, suppose lower and upper count contributions satisfy $L_h(z) \le C_j(z) \le U_h(z)$ for every member j of group h, on the same valid domain (including extended $+\infty$ values outside domains of influence). Then

$$\min\{C_0, \min_h L_h\} \;\le\; \min\{C_0, \min_j C_j\} \;\le\; \min\{C_0, \min_h U_h\}. \tag{11}$$

This follows from monotonicity of minimum. It does not supply L_h, U_h for coupled unknown trajectories, sampled births or receiving interactions. Count inequalities cannot simply be differentiated into pointwise flow inequalities. If uncertainty straddles active/inactive configurations, a worst-case penalty does not certify exact inactive behavior for the actual reference state.

Let $I(t)$ be cumulative offered population, including initial population, and $E(t)$ cumulative final exits, in vehicle equivalents. With all entrance, ramp and road waiting counted, system inventory is $I(t) - E(t)$. For common I, finite-horizon total system time and its error satisfy

$$J_T = \int_0^T [I(t) - E(t)]\,\mathrm{d}t, \tag{12}$$

$$|J_T - \widehat{J}_T| \le \int_0^T |E(t) - \widehat{E}(t)|\,\mathrm{d}t. \tag{13}$$

Units are vehicle-equivalent seconds. Full clearance requires certified tails for both models. A genuine exit-count enclosure gives a corresponding TST enclosure by integration; bin widths alone do not provide such an enclosure. Even a small objective error does not prove correct or useful gradients.

Disposition

Exact identical-signature grouping is a conditional equivalence rule. Broader compact reconstruction remains unqualified. The sampled task, point merge, priority, public data and TST are unchanged. The concrete grouping/approximation decision S3-G2-Q-G is open; no candidate is adopted and Stage 4 is unopened.

Source locations

Supplied Laval–Daganzo (2006), Section 2.2 and Appendix A, supplies the motion and descriptor semantics. Daganzo–Laval RR-2003-2, Sections 2–4/Appendix, and Simoni–Claudel, arXiv:1701.01026, Sections 2–3, support count/trajectory coupling under their stated laws. Equations here are conditional reviewer deductions, not a transferred theorem for the complete ramp task.